

ESPECIFICAÇÃO FORMAL DA LINGUAGEM MINI-LANG (BNF)

Camila Vanessa de Matos Sousa, Dalton Gomes Lobato,
Pedro Rafael Pereira de Oliveira, Vinícius Inácio dos Santos

1	<program>	→ <statement_list>
2	<statement_list>	→ <statement> <statement_list> ε
3	<block>	→ "{" <statement_list> "}"
4		
5	<statement>	→ <variable-decl> ";" <assignment> ";" <print-statement> ";"
6		<if-statement> <while-statement> <return-statement> ";"
7		<function-decl>
8		
9	<function-decl>	→ "def" TK_ID "(" <opt_formal_params> ")" ":" <type> <block>
10	<opt_formal_params>	→ <formal-params> ε
11	<formal-params>	→ <formal-param> <formal_param_tail>
12	<formal_param_tail>	→ "," <formal-param> <formal_param_tail> ε
13	<formal-param>	→ TK_ID ":" <type>
14		
15	<while-statement>	→ "while" "(" <expression> ")" <block>
16	<if-statement>	→ "if" "(" <expression> ")" <block> <opt_else>
17	<opt_else>	→ "else" <block> ε
18	<return-statement>	→ "return" <expression>
19	<print-statement>	→ "print" <expression> ";"
20		
21	<type>	→ "int" "real" "bool" "void"
22	<variable-decl>	→ "var" TK_ID ":" <type> "=" <expression>
23	<assignment>	→ "set" TK_ID "=" <expression>
24		
25	<expression>	→ <simple-expression> <relational_tail>
26	<relational_tail>	→ <relational-op> <simple-expression> <relational_tail> ε
27	<simple-expression>	→ <term> <additive_tail>
28	<additive_tail>	→ <additive-op> <term> <additive_tail> ε
29	<term>	→ <factor> <multiplicative_tail>
30	<multiplicative_tail>	→ <multiplicative-op> <factor> <multiplicative_tail> ε
31		
32	<factor>	→ TK_LIT_INT TK_LIT_REAL TK_LIT_TRUE TK_LIT_FALSE
33		TK_LIT_STRING TK_ID <id_tail> "(" <expression> ")"
34		<unary_op> <factor>
35		
36	<id_tail>	→ "(" <opt_actual_params> ")" ε
37	<opt_actual_params>	→ <actual-params> ε
38	<actual-params>	→ <expression> <actual_params_tail>
39	<actual_params_tail>	→ "," <expression> <actual_params_tail> ε
40		
41	<relational-op>	→ "<" ">" "==" "!=" "<=" ">="
42	<additive-op>	→ "+" "-" "or"
43	<multiplicative-op>	→ "*" "/" "and"
44	<unary_op>	→ "+" "-" "not"